

The Parameter Count of the Rope Medium

One Dimensionless Coupling, a Lorentz-Fixed Scale, and Why Symmetry Cannot Fix the Coupling

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Abstract

The classical rope theory rests on three microscopic primitives: the strand tension T , the bond stiffness κ , and the lattice spacing a . This paper asks how many of them are genuinely free, and answers the question completely. Three results follow, and we give the positive reduction and the negative impossibility equal weight. First, by dimensional analysis the three primitives form exactly ONE dimensionless group, $\Pi = \kappa a / T$; the other two degrees of freedom are pure unit conventions, so the theory has a single dimensionless coupling, not three. Second, requiring an emergent Lorentz-invariant signal speed fixes the line density $\mu = T/c^2$ — a genuine elimination of a further scale — but leaves Π untouched, because μ is dimensionally orthogonal to Π (it enters the dimensionless group with exponent zero). Third, and most usefully, we show that the value of Π cannot be fixed by dimensional analysis, Lorentz invariance, or lattice-dispersion consistency: the first is silent on values by construction, the second is orthogonal to Π , and the third bounds the absolute scale a (Lorentz violation $\sim (k a)^2/24$) without touching the dimensionless coupling. The value of Π is therefore a genuine free parameter, fixable only by a deeper theory or by matching one measured number. This converts ‘three unexplained constants’ into ‘one irreducible dimensionless parameter, with a proof of what cannot determine it.’ The dimensional and dispersion computations are reproducible via benchmarks/micromech/parameter_count.py.

What is established, and what is proven impossible

ESTABLISHED (positive): the primitives $\{T, \kappa, a\}$ carry exactly one dimensionless combination $\Pi = \kappa a / T$; two of the three are unit conventions. Lorentz invariance additionally fixes the line density $\mu = T/c^2$.

PROVEN IMPOSSIBLE (negative): the VALUE of Π cannot be determined by dimensional analysis, Lorentz invariance, or absence of lattice Lorentz violation. Π is a genuine free parameter; its value requires a deeper theory or one empirical anchor. This is a limit of these methods, stated as a result, not a gap left unmentioned.

1. The Three Primitives

Every quantity in the classical rope theory is built from three microscopic inputs:

- T — the strand tension, with dimension of force, $[T] = M L T^{-2}$.

- κ — the bond stiffness, fixed so that the locking energy $J = T^2/\kappa$ is an energy, giving $[\kappa] = M Ti^{-2}$.
- a — the lattice spacing, $[a] = L$.

These are primitive parameters. That is entirely acceptable for an effective theory — every effective theory has undetermined parameters at its cutoff — but it is the deepest remaining arbitrariness in the programme, and it is worth establishing exactly how much of it is real.

2. Result 1: Three Primitives, One Dimensionless Coupling

The physical content of a set of dimensionful parameters is carried entirely by their independent dimensionless combinations (the Buckingham π theorem). Forming the dimension matrix of $\{T, \kappa, a\}$ in the base dimensions (M, L, Ti),

$$[T \ \kappa \ a] = M:(1, 1, 0) \quad L:(1, 0, 1) \quad Ti:(-2, -2, 0) \quad (2.1)$$

gives a matrix of rank 2 on three columns. By Buckingham's theorem the number of independent dimensionless groups is $3 - 2 = 1$. Computing the null space, that single group is

$$\Pi = \kappa a / T = T a / J \quad (2.2)$$

(using $J = T^2/\kappa$). Physically Π compares the work to displace a strand by one lattice spacing, $T a$, against the locking energy J that resists it. The remaining two degrees of freedom among $\{T, \kappa, a\}$ are pure unit conventions: with the mass, length, and time scales free, two of the three primitives can be set to unity, and every physical prediction depends on them only through Π .

Result 1

The rope theory has ONE dimensionless coupling, $\Pi = \kappa a / T$, not three independent constants. Two of the three primitives are unit conventions. This is the honest reduction of the parameter count — Derived, and reproducible.

3. Result 2: Lorentz Invariance Fixes the Scale μ , Not the Coupling Π

A natural hope is that requiring the rope medium to respect Lorentz invariance — a single universal signal speed c — might fix Π . We test this directly, and it does something real but not that.

3.1 μ is a fourth, independent primitive

The wave speed on the medium is $c = \sqrt{T/\mu}$, where μ is the line mass density, $[\mu] = M L^{-1}$. A short argument shows μ cannot be built from $\{T, \kappa, a\}$: both T and κ carry time-dimension Ti^{-2} and nothing else in time, so any product $T^x \kappa^y$ has time-dimension $Ti^{-2(x+y)}$; a time-independent quantity like μ forces $x + y = 0$, which then forces mass-dimension $x + y = 0 \neq 1$. The density μ is therefore dimensionally independent of the three primitives — a fourth input.

3.2 Lorentz invariance fixes μ but is orthogonal to Π

Emergent Lorentz invariance imposes one equation, $c^2 = T/\mu$, i.e.

$$\mu = T / c^2 , \quad (3.1)$$

which genuinely eliminates μ as a free scale. But adding μ to the primitive set and recomputing the dimensionless groups gives, again, exactly one group — and μ enters it with exponent zero:

$$[T \kappa a \mu] \rightarrow \text{single group } \Pi = \kappa a / T \quad (\mu\text{-exponent} = 0). \quad (3.2)$$

Because μ does not appear in Π , fixing μ via (3.1) says nothing whatever about Π . Lorentz invariance sets a scale; it is dimensionally orthogonal to the dimensionless coupling and cannot constrain its value.

Result 2

Lorentz invariance fixes the line density $\mu = T/c^2$ — a real elimination of a scale — but leaves the dimensionless coupling $\Pi = \kappa a / T$ entirely unconstrained, because μ is dimensionally orthogonal to Π (enters the dimensionless group with exponent zero). Derived.

4. Result 3: The Value of Π Cannot Be Fixed by These Symmetries

The most useful outcome is the sharp statement of what cannot be done, which turns an open hope into a settled result. Three candidate routes to the VALUE of Π are ruled out.

4.1 Dimensional analysis is silent on values

By construction, dimensional analysis identifies which dimensionless combinations exist; it never assigns them numerical values. It reduces $\{T, \kappa, a\}$ to Π (Result 1) but cannot say whether Π is 10^{-3} or 10^3 .

4.2 Lorentz invariance is orthogonal to Π

By Result 2, the one equation Lorentz invariance supplies fixes μ , a quantity orthogonal to Π . It therefore cannot fix Π 's value.

4.3 Absence of Lorentz violation bounds the scale a , not Π

One might hope the requirement that no Lorentz violation be observed constrains Π . The lattice dispersion is $\omega^2 = c^2 (2/a^2)(1 - \cos k a) = c^2 k^2 (1 - (k a)^2/12 + \dots)$, so the relative Lorentz violation at wavenumber k is

$$| \omega - c k | / (c k) \approx (k a)^2 / 24 , \quad (4.1)$$

confirmed numerically (values 1.04×10^{-2} , 4.17×10^{-3} , 1.04×10^{-3} at $k a = 0.2, 0.1, 0.05$, matching the prediction to better than 1%). This vanishes as $a \rightarrow 0$, so the constraint is that the lattice spacing a must be small in ABSOLUTE units — fine enough to hide Lorentz violation at tested energies. That bounds the dimensionful scale a ; it says nothing about the dimensionless $\Pi = \kappa a / T$, in which a is measured against T and κ , not against an external length.

Result 3 (the impossibility)

The VALUE of $\Pi = \kappa a / T$ cannot be determined by dimensional analysis (silent on values), by Lorentz invariance (orthogonal to Π), or by absence of lattice Lorentz violation (bounds the absolute scale a , not the dimensionless Π). Π is a genuine free parameter of the theory. Its value can come only from a deeper

theory beneath the ropes, or from matching a single measured quantity, after which all else is a prediction relative to that anchor. Derived (impossibility within the stated methods).

5. The Parameter Count, Summarised

Quantity	Status after this analysis
T, κ, a (three primitives)	Two are unit conventions; one physical combination remains
$\Pi = \kappa a / T$	The single dimensionless coupling — value undetermined (free parameter)
μ (line density)	Fixed by Lorentz invariance: $\mu = T/c^2$
Value of Π	Provably not fixable by dimensional analysis, Lorentz, or dispersion

The net movement is real: from three unexplained dimensionful primitives to one irreducible dimensionless coupling, one Lorentz-fixed scale, and a proof of what cannot pin the coupling. The theory's genuine freedom is a single number.

6. Status of Each Result

Result	Status
3 primitives $\rightarrow$ 1 dimensionless coupling $\Pi = \kappa a / T$	Derived — Buckingham π ; reproducible
μ independent of $\{T, \kappa, a\}$; Lorentz fixes $\mu = T/c^2$	Derived — dimensional argument
μ orthogonal to Π (exponent 0 in the group)	Derived — null-space computation
Lorentz violation $\sim (k a)^2/24$ bounds scale a	Derived — dispersion; numerically confirmed
Value of Π not fixable by these symmetries	Derived — impossibility within stated methods
Value of Π from a deeper theory	Open — outside these methods; not claimed

Conclusion

Asked whether any of the rope theory's primitive constants can be eliminated, the answer is a precise one. Two of the three primitives are unit conventions, leaving a single dimensionless coupling $\Pi = \kappa a / T$; Lorentz invariance eliminates a further scale by fixing the line density $\mu = T/c^2$; and the value of Π cannot be determined by dimensional analysis, Lorentz invariance, or lattice-dispersion consistency — a fact we prove rather than leave open. The programme's irreducible freedom is thus exactly one number, whose value must come from a deeper theory or from a single empirical anchor. This is not a derivation of the constants from nothing, which would be the same wall all of fundamental physics faces; it is the honest and useful thing that can be

established — a clean parameter count, a scale fixed by symmetry, and a proof of where the remaining wall stands and why symmetry alone does not breach it.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **FND-005 [Derived]:** Parameter count [benchmark: benchmarks/micromech/parameter_count.py]

- **FND-006 [Derived]:** Lorentz invariance fixes $\mu = T/c^2$ but is orthogonal to Π ; value of Π not fixable by symmetry [benchmark: benchmarks/micromech/parameter_count.py]

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.